A Pentagonal-Drum Path

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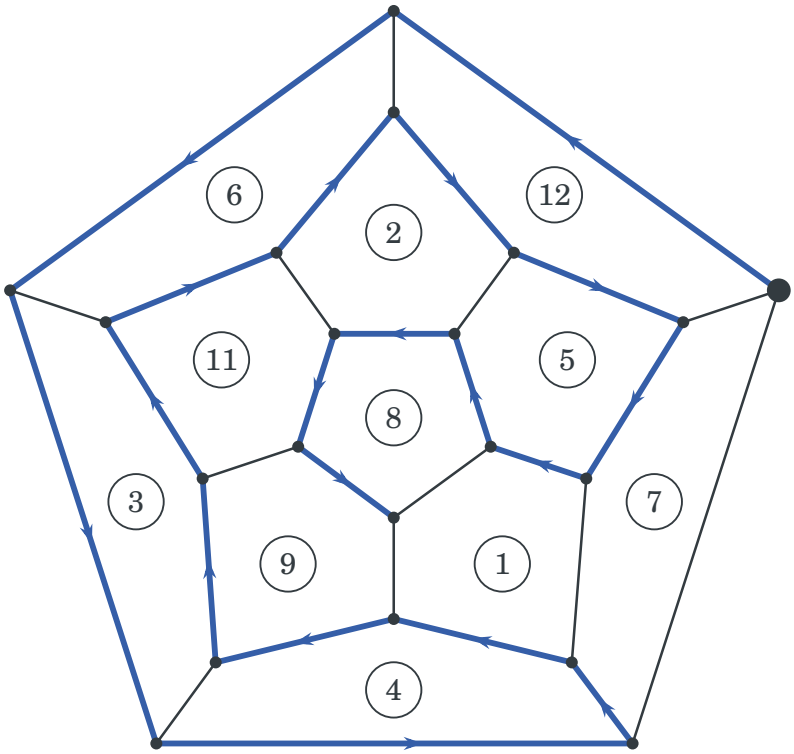

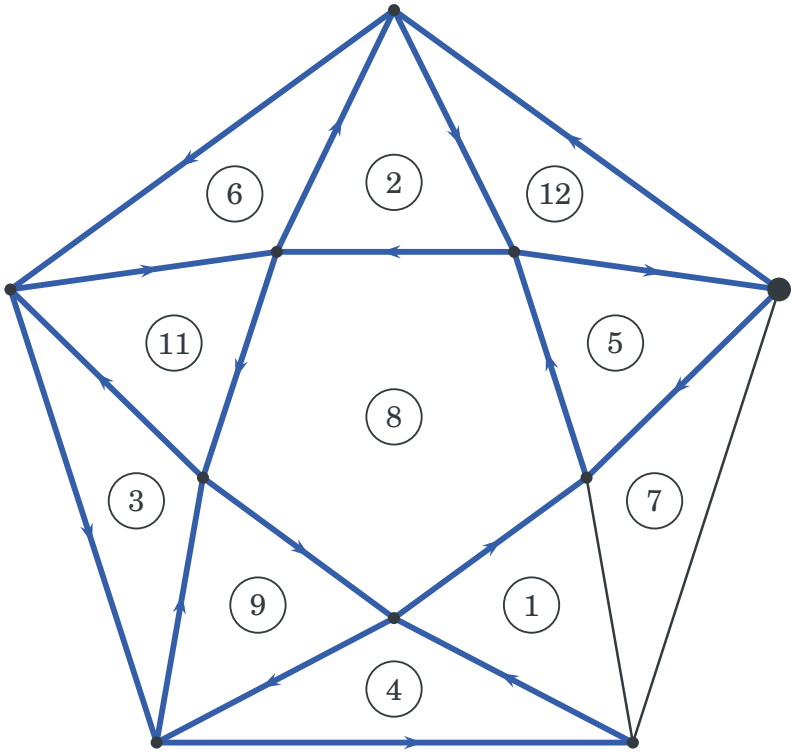

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Original Hamiltonian path

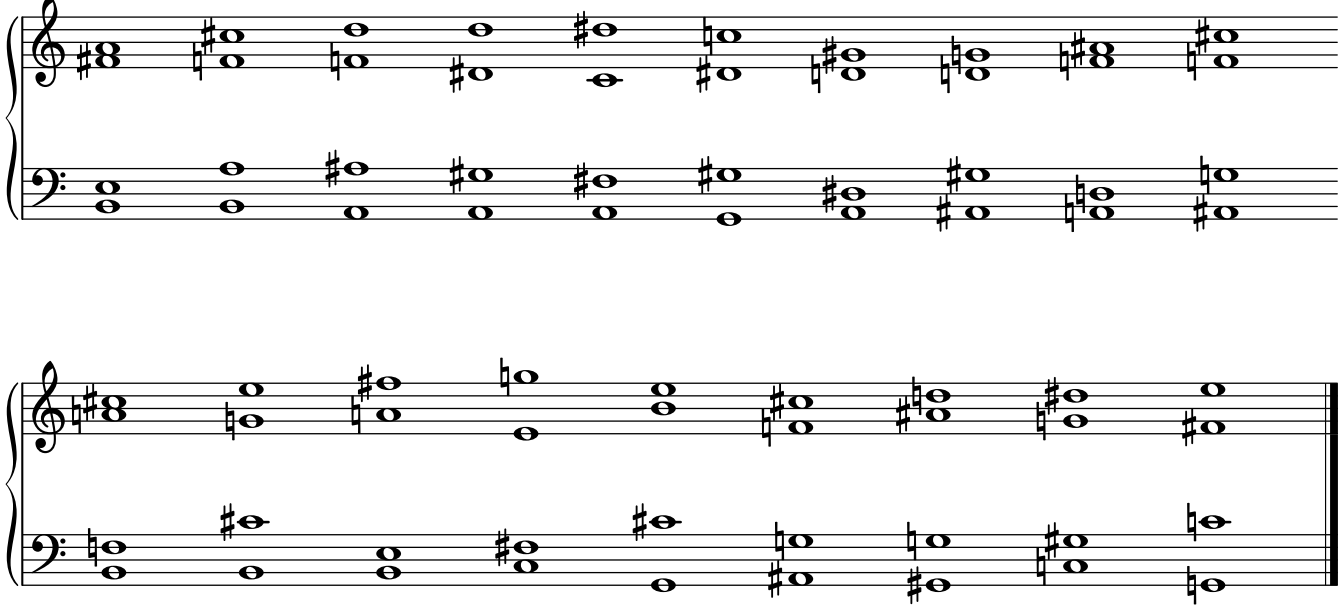
Each face of the dodecahedron carries a chromatic pitch class. The three faces meeting at a vertex form one chord. The marked path visits all twenty vertices once, starting at the heavy dot.
The absent number 10 denotes the outside face.

Twelve-tone equal temperament;
A4 = 440 Hz.

1: C	2: C♯	3: D	4: D♯
5: E	6: F	7: F♯	8: G
9: G♯	10: A	11: A♯	12: B

Pentagonal drum

Contract the five outer spokes and the five spokes around the central pentagon. The vertices merge in pairs; four faces meet at each new vertex, giving a four-note chord.

After the contraction, immediate repetitions are removed and the route is extended by its last inner edge. The nineteen chords follow Figure 15 and Example 2 of the paper.

Immobile